Dictionary Exercise

You are developing a simple Student Grade Manager for a university. The program stores student names and grades. Each student has a unique name associated with one grade. Create a C# Console Application using Dictionary<string, int>.

1. Create a Dictionary

- Create a Dictionary<string, int> called studentGrades.
- Add Peter → 10, Sara → 12, and Henrik → 7 using Add().

2. Display All Students

- Use foreach to display each student's name and grade.
- Use Key for the name and Value for the grade.
- Print the total number of students using Count.

3. Update and Add Students

- Change Peter's grade from 10 to 12 using his key.
- Add Emma with grade 10.

4. Search for a Student

- Use ContainsKey() to check whether Sara exists.
- If Sara exists, print her grade; otherwise print 'Student not found'.

5. Remove a Student

- Remove Henrik using Remove().
- Display the remaining students and grades using foreach.
- Print the final number of students.

Starter Code

```
using System;
using System.Collections.Generic;

public class Program
{
    public static void Main(string[] args)
    {
        // TODO 1: Create Dictionary<string, int>
        // TODO 2: Add Peter, Sara, and Henrik
        // TODO 3: Display all students using foreach
        // TODO 4: Print total number of students
        // TODO 5: Update Peter's grade to 12
        // TODO 6: Add Emma with grade 10
        // TODO 7: Check whether Sara exists and print her grade
        // TODO 8: Remove Henrik
        // TODO 9: Display final dictionary and count
    }
}
```

Expected Output

```
Initial Student Grades:
Peter: 10
Sara: 12
Henrik: 7
Total students: 3
```

```
Sara's grade: 12

Final Student Grades:
Peter: 12
Sara: 12
Emma: 10
Total students: 3
```

Note: The order of students in the output may vary.